

**BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
WORK INTEGRATED LEARNING PROGRAMMES**

Part A: Course Design

Course Title	Data Management for Machine Learning
Course No(s)	DSE* ZG529 / AIML* ZG529
Credit Units	4
Content Authors	Pravin Y Pawar
Version	1.1

Course Description

Data Models and Query Languages: Relational, Object-Relational, NoSQL data models; Declarative (SQL) and Imperative (MapReduce) Querying; Data Encoding: Evolution, Formats, Models of dataflow; Machine learning workflow; Data management challenges in ML workflow; Data Pipelines and patterns; Data Pipeline Stages: Data extraction, ingestion, cleaning, wrangling, versioning, transformation, exploration, feature management; Modern Data Infrastructure: Diverse data sources, Cloud data warehouses and lakes, Data Ingestion tools, Data transformation and modelling tools, Workflow orchestration platforms; ML model metadata and Registry, ML Observability, Data privacy and anonymity.

Course Objectives

The course aims at providing:	
CO1	Introduction to the data models, storages and querying languages used in data management emphasizing on machine learning aspects
CO2	Required guidance on architecture of modern data platform, usage and types of data pipelines
CO3	Hands-on exposure to the common techniques, and tools used by data engineers to support build, test, deploy and automate the machine learning pipelines
CO4	Exposure to the industry best practices essential to deal with data privacy, metadata and observability

Text Book(s)

T1	Fundamentals of Data Engineering: Plan and Build Robust Data Systems by Reis and Housley
T2	Reliable Machine Learning By Cathy Chen, Niall Richard Murphy, Kranti Parisa, D. Sculley, Todd Underwood

Reference Book(s) & other resources

R1	Designing Data-Intensive Applications by Martin Kleppmann
R2	Data Pipelines Pocket Reference by Densore
R3	Building Machine Learning Pipelines by Hapke, Nelson

Learning Outcomes:

Students will be able to :	
LO1	Understand the necessity, position and role of data management components appearing in the modern data stacks
LO2	Acknowledge the patterns, challenges and possible solutions associated with the data ingestion, flow, storage and processing on data platforms
LO3	Gain experience in designing and handling the dataflow during machine learning pipeline by means of state-of-art tools
LO4	Apply the acquired conceptual data management knowledge and practices over a real-world machine learning workflow addressing the model metadata, privacy and monitoring aspects

Part B: Course Handout

Academic Term	II Semester 2022-2023
Course Title	Data Management for Machine Learning
Course No	DSE* ZG529 / AIML* ZG529
Lead Instructor	Pravin Y Pawar

Glossary of Terms

Module	M	Module is a standalone quantum of designed content. A typical course is delivered using a string of modules. M2 means module 2.
Contact Hour	CH	Contact Hour (CH) stands for an hour long live session with students conducted either in a physical classroom or enabled through technology. In this model of instruction, instructor led sessions will be for 32 CH.
Recorded Lecture	RL	RL stands for Recorded Lecture or Recorded Lesson. It is presented to the student through an online portal. A given RL unfolds as a sequences of video segments interleaved with exercises.
Lab Exercises	LE	Lab exercises associated with various modules
Self-Study	SS	Specific content assigned for self-study
Homework	HW	Specific problems/design/lab exercises assigned as homework

Modular Structure

Module Summary

No.	Content of the Module
M1	Foundations of data management
M2	Modern Data Platform
M3	Data Management in ML Workflow
M4	Advanced Topic in Data Management

Detailed Structure

M1: Foundations of data management

Contact Session 1-2

Session	Type	Description/Plan	Reference
1	CH1	- Data Management Principles - Data Management Components	T2
	CH2		
2	CH3	- Data Models and Query Languages - Data Encoding	R1 T1
	CH4		
Post CS	LE	Lab 1	

M2: Modern Data Platform

Contact Session 3-4

Session	Type	Description/Plan	Reference
3	CH5	- Data Architectures - Modern Data Stack - Data Pipelines and patterns	T1
	CH6		
4	CH7	- Data Storage - Data Science Infrastructure - Serving Data for Analytics and ML	T1
	CH8		
Post CS	LE	Lab 2	

M3: Data Management in ML Workflow

Contact Session 5-12

Session	Type	Description/Plan	Reference
5	CH9	ML Workflow/lifecycle	T2 R3
	CH10	• Data Pipeline vs ML Pipeline • ML Pipeline Stages • Training / Serving pipeline • Data management challenges in ML workflow	
6	CH11	Data Collection / Ingestion	T1
	CH12	• Diverse data sources • Data generation in source systems • Batch Ingestion • Message and Stream Ingestion • Ingestion strategies	
7	CH13	Data Validation	R3
	CH14	• Common problems with data • Data skew and drift • Bias and Fairness • Data leakage • Data validation approaches	
8	CH15	Analytics Engineering	Instructor-supplied material
	CH16	• Data Integration • Data Transformation • Data Partitioning • Data Versioning • Test data management challenges	
9	CH17 CH18	Data Analysis • Types of Analytics • Data Exploration and Visualizations • Data Cubes and OLAP • Data Cube Operations • Data Cubes and ML	Instructor-supplied material
10	CH19	Feature Preparations	T2
	CH20	• Feature life cycle • Data Annotation / labeling • Data augmentation and Data Synthesis • Common Feature Engineering Operations	



		- Feature Importance - Feature Generalization - Feature Stores	
11-12	CH21 CH22	ML Experimentation & Metadata	Instructor-supplied material
	CH23 CH24	- Model training & experimentation - Model Analysis & Validation - ML Metadata Store - Dataset, Feature, Label, Pipeline metadata - ML Experiment Tracking data - ML model metadata and Registry	
Post CS	LE	Lab 3, 4	

M4: Advanced Topic in Data Management

Contact Session 14-16

Session	Type	Description/Plan	Reference
13	CH25 CH26	Distributed Data Processing - Big Data Analytics - Technologies for big data processing - Distributed and Parallel data processing - In-memory data processing - Hadoop, Spark, Kafka as exemplar architecture	
14	CH27 CH28	Data Privacy and anonymity - Data privacy issues - Differential privacy - Anonymization - Methods to preserve privacy - Federated learning - Encrypted ML	T2 Instructor-supplied material
15	CH29 CH30	Data Observability - Data Observability - Data downtime - Five pillars - Tools selection	T2 Instructor-supplied material
16	CH31 CH32	ML Monitoring & Observability - Causes of ML System failure - Data Distribution Shifts - Problems with ML Production Monitoring - ML-specific metric	T2 Instructor-supplied material



		- Monitoring Toolbox - Monitoring vs Observability	
Post CS	SS	To be identified	

Experiential Learning Component

Lab	Topic	
1	Design and implement the simple data flows involving various data formats Modes of data flows a) Through Databases – use SQL / Custom Program to read/write into databases b) Through REST/RPC – Synchronous mechanism for data exchange c) Through Message Brokers / Queues – Asynchronous mechanism for data exchange	Virtual Labs
2	Build a Modern Data Stack Components a) a fully managed ELT data pipeline b) a cloud-based columnar warehouse or data lake as a destination c) a data transformation tool d) A business intelligence or data visualization platform.	Virtual Labs
3	Manage Machine Learning Model Metadata using MLFlow / Neptune Components a) Projects b) Experiments c) Model metadata d) Model tracking / logging e) Model Registry	Virtual Labs
4	Construct a Machine Learning Pipeline with Data Versioning Tool Components a) Data Pipeline b) Data Versioning Tool c) Feature Store d) ML Pipeline e) Prediction Service	Virtual Labs

Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

No	Name	Type	Duration	Weight	Day, Date, Session, Time
EC-1	Experiential learning Assignment / Quiz -I	Take Home	15 days	10%	TBA
	Experiential learning Assignment-II	Take Home	15 days	20%	TBA
EC-2	Mid-Semester Test	Closed Book	2 hours	30%	Per programme schedule
EC-3	Comprehensive Exam	Open Book	3 hours	40%	Per programme schedule

Syllabus for Mid-Semester Test (Closed Book): Topics in Session Nos. 1 to 7

Syllabus for Comprehensive Exam (Open Book): All topics (Session Nos. 1 to 16)

Important links and information:

Elearn portal: <https://elearn.bits-pilani.ac.in>

Students are expected to visit the Elearn portal on a regular basis and stay up to date with the latest announcements and deadlines.

Contact sessions: Students should attend the online lectures as per the schedule provided on the Elearn portal.

Evaluation Guidelines:

1. EC1 consists of two assignments. Announcements will be made available on the portal, in a timely manner.
2. For Closed Book tests: No books or reference material of any kind will be permitted.
3. For Open Book exams: Use of books and any printed / written reference material (filed or bound) is permitted. However, loose sheets of paper will not be allowed. Use of calculators is permitted in all exams. Laptops/Mobiles of any kind are not allowed. Exchange of any material is not allowed.
4. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam which will be made available on the Elearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the online lectures, and take all the prescribed evaluation components such as Assignment/Quiz, Mid-Semester Test and Comprehensive Exam according to the evaluation scheme provided in the handout.